ARTIFICIAL INTELLIGENCE (3170716) IMP Question Bank

Sr.No.	Questions	Marks	CO
1	**Define the term: Artificial Intelligence. State various applications of Artificial Intelligence.**	3	
2	Enlist and briefly discuss the major task domains of Artificial Intelligence.	4	
3	**Explain AI problem characteristics in detail.**	7	
4	Explain the “Turing test”.	3	
5	**Differentiate between DFS and BFS.**	**3**	
6	What is a “control strategy” and what are its characteristics?	4	
8	State Monkey and Banana problem. Give its state space representation.	7	
9	**Consider the water jug problem stated below:** **You are given two jugs, a 5-litre one and a 4-litre one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2-litre of water into the 4-litre jug?** **Explain how this problem can be solved using State Space Search. Also, give the production rules to solve this problem and derive a feasible solution using the same.**	7	
10	In the missionaries and cannibals problem, three missionaries and three cannibals must cross a river using a boat which can carry at most two people, under the constraint that, number of cannibals should be lesser than or equal to the missionaries on either side. The boat cannot cross the river by itself with no people on board. For the above mentioned problem, describe state space representation, actions, start and end state.	7	
11	**Discuss and Analyze Tower of Hanoi problem with respect to the seven problem characteristics**	7	
12	What is State Space of a Problem?	3	
13	What is production system? Discuss the component of production system	4	
14	**Explain the state space search with the use of 8 Puzzle Problem**	7	CO1
15	What is heuristic search? Discuss with an example	3	
16	Trace the constraint satisfaction procedure solving the following cryptarithmetic problem. R I N G + D O O R ----------------------------------- B E L L	7	
17	Explain Hill Climbing method in brief.	3	
18	**What is A* search? Describe the different stages of A* search with an example.**	7	
19	Write an objective of hill climbing method. Discuss any one type of hill climbing in detail with its algorithm.	7	
20	Write the proposed solution(s) for the problems occurred in hill climbing.	3	
21	Explain problem reduction using “AND-OR” graph.	3	
22	**Solve the following cryptarithmetic problem using constraint satisfaction.** **L E O + L E E = A L L**	4	
23	Define constraint satisfaction problem (CSP). How CSP is formulated as a search problem?	7	
24	Explain local maxima, plateau, and ridge in detail.	4	
25	Explain mean-end analysis approach to solve AI problems.	4	
26	Define i) Local Maximum ii) Plateau iii) Ridge	3	
27	Explain different approaches to knowledge representation.	7	
28	**Explain Inference Rules in Propositional Calculus.**	4	
30	Explain the various method of knowledge representation with suitable example	7	
31	Consider following sentences: • John likes all kinds of food. • Apples are food. • Chicken is food. • Anything anyone eats and isn’t killed by is food. • Bill eats peanuts and is still alive. • John eats everything Bill eats. (1)Translate these sentences into formulas in predicate logic. (2)Prove that John likes peanuts using Backward chaining.	7	

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32	Describe the steps involved in unification in predicate logic.	3	CO2
33	**Translate these sentences into formulas in predicate logic. 1. John likes all kinds of food. 2. Apples are food. 3. Chicken is food. 4. Anything anyone eats and isn't killed-by is food. 5. Bill eats peanuts and is still alive. 6. Sue eats everything Bill eats.**	3	
34	Consider the following facts: (1) Steve only likes easy courses. (2) Science courses are hard. (3) All the courses in the HaveFun department are easy. (4) BK301 is a HaveFun department course. Use resolution to answer the question "What course would Steve like?"	7	
35	**Differentiate prepositional & predicate logic.**	3	
36	What is clausal form? How is it useful?	4	
37	Write unification algorithm and explain resolution in predicate logic.	7	
38	Explain the steps of unification in predicate logic.	4	
39	What are the limitations of Propositional Logic?	4	
40	Consider the following sentences: • Raj likes all kinds of food. • Apples are food. • Anything anyone eats and isn't killed by is food. • Sachin eats peanuts and is still alive. • Vinod eats everything Sachin eats. Now, attempt following: i. Translate these sentences into formulas in predicate logic. ii. Use resolution to answer the question, "What food does Vinod eat?"	7	
41	Write and explain algorithm for resolution in propositional logic with suitable example	5	
42	Consider the following sentences: • Rita likes all kinds of food. • Apples are food. • Anything anyone eats and isn't killed by is food. • Rahi eats peanuts and is still alive. • Tanvi eats everything Rahi eats. i. Translate these sentences into formulas in predicate logic. ii. Use resolution to answer the question, "What food does Tanvi eat?"	7	
43	Discuss forward reasoning in brief.	4	
44	Briefly discuss Declarative and procedural knowledge.	3	
45	**Explain forward and backward reasoning with suitable example(s).**	7	
46	**Write the basic differences between declarative and procedural knowledge.**	3	
47	Differentiate forward chaining and backward chaining with suitable example.	7	
49	Differentiate procedural and declarative knowledge	3	
51	Define the following. 1. Modus Ponens 2. Horn Clause 3. Existential Quantifier	3	
58	Write a short note on non monotonic reasoning	4	
62	State the Bayes theorem. Illustrate how a Bayesian Network can be used to represent causality relationship among attributes.	7	
64	A bag I contains 4 white and 6 black balls while another Bag II contains 4 white and 3 black balls. One ball is drawn at random from one of the bags, and it is found to be black. Find the probability that it was drawn from Bag I.	4	
65	Define Frames. Draw Semantic Net for following statements. a) Every kid likes candy. b) Every school going kid likes candy.	7	

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66	Explain Semantic Net & Frame with suitable example.	7	
67	**Explain Minimax search procedure with suitable example.**	7	CO3
68	**Explain the Alpha-Beta Cutoffs Procedure in Game Playing.**	7	
69	**Simulate the working of Tic-tac-toe problem with Minimax technique.**	7	
70	Explain Hierarchical planning in brief.	3	
71	Discuss goal stack planning in brief.	4	
72	Explain Components of a Planning System.	3	
73	Explain how planning is different from search procedure?	3	
74	Explain syntax and semantic analysis phases of Natural Language Processing.	4	CO4
75	What is Natural Language Processing? Discuss applications of NLP in brief.	3	
76	Define Natural language processing and explain Discourse and Pragmatic processing.	7	
77	State the factors which may make understanding of natural language difficult for a computer.	3	
78	Explain morphological and syntax analysis phases of NLP.	3	
79	What is learning? Explain various learning techniques.	4	
80	**What is Artificial neural network? Explain with example.**	4	
81	**Discuss Hopfield network in brief.**	4	
82	Discuss different types of learning in artificial neural networks. Also write the differences between/among them.	7	
83	Explain connectionist models. What is perceptron?	3	
84	**Explain the algorithm for Backpropagation in Neural Networks.**	**4**	
85	**Explain supervised and unsupervised learning**	**3**	
86	Explain knowledge acquisition in expert system.	4	
87	Describe the Expert System Development Procedure.	7	
88	Explain the architecture of expert system with suitable sketch	7	
89	Explain Fuzzy Logic.	3	
90	Discuss the following terms with respect to fuzzy logic theory: - Membership function - Linguistic variable	4	
91	Explain different defuzzification methods.	3	
92	Discuss the termination parameters used in genetic algorithms in detail.	7	
93	Explain various genetic operators used in genetic algorithms in brief.	7	
94	Explain Significance of the Genetic Operators.	4	
95	Compare Fuzzy Vs Crisp Logic.	3	
96	Explain Various Types of Cross Over Operators in Genetic Algorithm.	7	
97	Explain Roulette-Wheel selection method of genetic algorithm.	7	
98	Describe the phases of genetic algorithm.	7	
100	Write a prolog program to find minimum number from the given input list.	7	CO5
101	Write Prolog programs to perform the following with LIST: i) Delete an element from a list ii) Concatenating of two lists	7	
102	Write a prolog program to count the number of elements present in the given list.	4	
103	**Demonstrate the use of Cut and Fail Predicates in Prolog with example.**	**7**	
104	**Write following prolog programs:** **1. To find factorial of a given number** **2. To find the nth element of a given list**	**7**	
106	**Write a Prolog program which contains three predicates: male, female, parent. Make rules for following family relations: father, mother, grandfather, grandmother, brother, sister.**	**7**	